A Nonlinear Weighted Graph Convolution Network Based on Manifold Geometric Regularization for Hyperspectral Image Classification

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***Abstract*— Extracting spatial–spectral joint features has become a critical approach for improving model classification performance in the field of hyperspectral image classification (HSIC). However, existing methods fail to fully exploit nonlinear spatial–spectral information. Unlike traditional convolutional neural networks (CNNs), graph convolutional networks (GCNs) can extract nonlinear spatial information. Nevertheless, both methods lack an accurate measurement of local neighborhood information, leading to blurred classification boundaries for ground objects. Additionally, the high-dimensional nature of hyperspectral data results in poor generalization and redundant information of trained models. To address these three issues, a nonlinear weighted graph convolution network based on manifold geometric regularization (MGR-NWGCN) method is devised for HSIC. Specifically, a nonlinear weighted graph convolution network (NWGCN) module is designed, which utilizes a graph-in-graph structure based on cosine similarity-based normalized weighted graph convolution to extract nonlinear spatial–spectral information. Then, the manifold curvature regularization (C-MGR) module is implemented to improve the accuracy of similarity measurement and to enhance the generalization ability of the model, which constrains the model to form flatter feature manifold surfaces. Finally, the manifold intrinsic dimensionality regularization (ID-MGR) module is developed with the aim of eliminating redundant information, which embeds noise onto the surface of a low-dimensional manifold. The superior classification performance and robustness of the proposed MGR-NWGCN method are validated through extensive experiments on four datasets, with comparisons conducted against nine methods.**

***Index Terms*— Graph convolutional network (GCN), hyperspectral image classification (HSIC), manifold geometric regularization (MGR), nonlinear spatial–spectral information.**

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I. Introduction

SPECTRAL signatures are obtained by measuring the amount of emitted, reflected, or transmitted light from observed materials, representing a wealth of distinct information [1], [2], [3]. A hyperspectral image (HSI) is composed of spectral and spatial signatures, which can be utilized in a wide range of applications. A brief list of these applications includes research in the field of solid Earth, resource management, and fine-grained classification of crops [4], [5], [6], [7]. HSI techniques have been widely applied in various domains, including HSI classification, target detection [8], and super-resolution [9]. HSI classification (HSIC) is a technique that classifies each pixel into different categories by taking advantage of the fine spectral resolution of HSIs [10], [11].

The inherent spectral nonlinearity of HSI data has been revealed in the spectral domain, which is primarily due to the multiple scattering between sun and observed materials and the heterogeneity of pixels [12]. To tackle the nonlinear spectral information, different methods based on kernel functions, such as nonlinear support vector machines (SVMs), kernel Fisher discriminant analysis (KFDA), and radial basis function (RBF), have been used for HSIC [13]. The kernel function of these methods maps data from the original input feature space to a higher dimensional feature space, making nonlinear data become linearly separable in this higher dimensional feature space. However, the problem with kernel-based methods is that the selection of parameters has a great impact on the classification results, and they require hyperspectral data to satisfy a specific nonlinear distribution [14], [15].

Manifold learning methods, unlike kernel function methods, can effectively explore the internal structure of data, and are not sensitive to data distribution, thus having stronger robustness [16]. Bachmann et al. [17] proposed a new algorithm based on manifold learning to exploit the nonlinear structure of HSI. The ability of manifold learning to recover low-dimensional nonlinear structure has been demonstrated in the machine-learning community [18].

All the methods mentioned above solely achieve the extraction of shallow nonlinear spectral features [19]. In contrast, deep learning (DL)-based models can capture more abstract features of HSI data through their multilayer structures [20], [21]. Chen et al. [22] proposed a feature extraction approach based on a convolutional neural network (CNN) to extract

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spatial–spectral features for HSIC. Lee and Kwon [23] proposed an approach using a multiscale convolutional filter bank to form a joint spatio-spectral feature map, achieving enhanced classification performance. Roy et al. [24] explored a hybrid 3-D–2-D CNN to extract joint spatial–spectral feature representation.

Nevertheless, CNNs may lose sequence structure when representing hyperspectral pixels [25]. Mou et al. [26] first proposed a recurrent neural network (RNN) framework for HSIC. Transformers have been proven effective for HSIC. Hong et al. [27] proposed a backbone network called SpectralFormer, which is capable of learning local sequence information. Sun et al. [28] proposed a spectral–spatial feature tokenization transformer method to represent abstract semantic features. Transformers can represent the sequence attributes of spectral signatures but result in information loss due to the incorporate average pooling. To solve this intractable obstacle, WaveFormer [29] is a novel transformer-based approach that leverages wavelet transforms for invertible downsampling.

All the aforementioned patch-based methods are capable of extracting local deep abstract features from HSI, yet they fail to take into account the characteristics of non-Euclidean HSI data. Non-Euclidean characteristics refer to the irregular spatial structures of the data [30], [31], [32]. Graph convolutional networks (GCNs) possess the excellent capability of representing the non-Euclidean data [33], [34]. Wan et al. [35] proposed a multiscale dynamic GCN (MDGCN), which can extract abstract nonlinear features from arbitrarily structured non-Euclidean data. MiniGCN has been developed in [36] to capture long-range spatial–spectral feature representations, while lowering computational cost. In [37], the attention mechanism called graph attention network (GAT) is introduced into the field of HSI processing, and the weighted feature fusion of CNNs and GAT achieves competitive results. However, due to the fixed neighborhood receptive field, the aforementioned models lack consideration of multiscale spatial information. To address this limitation, in [38], a multibranch graph neural network model (MBGNN) is developed to extract multiscale spatial information. Nevertheless, existing graph convolution-based methods can handle irregular spatial information effectively, but they are limited in processing boundary information. As a result, boundary pixels are often misclassified, especially when labeled samples are scarce [39].

The latest HSIC methods have recognized the importance of extracting joint spatial–spectral information [40], [41]. However, they still overlook the consideration of spatial nonlinearity and spectral nonlinearity. While DL is employed to extract deep abstract features, the key to achieving better classification performance in HSIC tasks lies in designing models that better adapt to the non-Euclidean spatial structure of HSI data and the manifold structure of the spectral feature space [42]. The nonlinearity of spatial and spectral information, as illustrated in Fig. 1, refers to two aspects: spatial nonlinearity refers to the nonlinear spatial structure of adjacent pixels, and spectral nonlinearity indicates the nonlinear distribution of features in high-dimensional space.

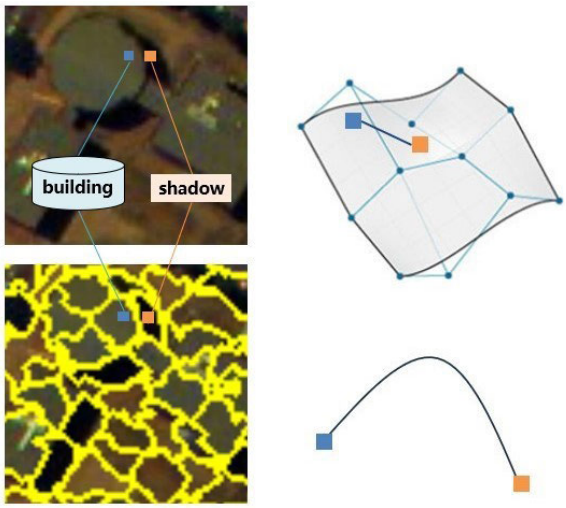


Fig. 1. (a) Spatial nonlinear information. (b) Spectral nonlinear information. As shown in (a), spatial nonlinearity refers to the positional relationship between buildings and shadows that cannot be simply described by planar geometric distances. As illustrated in (b), spectral nonlinearity indicates that the features are distributed on a manifold surface where distances cannot be accurately measured.

Existing methods need to consider not only the inherent nonlinear joint spatial–spectral structural information of HSI data, but also the spectral redundancy information. Fang et al. [43] employed soft band selection to suppress redundant bands in the coding phase. While emphasizing the bands containing richer information, generate discriminative feature vectors with greater noise robustness. Liao et al. [44] proposed a seed extend by entropy density algorithm to alleviate the seed point redundancy. However, the aforementioned methods for reducing hyperspectral redundancy, which linearly filter bands and seeds, may lead to the loss of important feature information in HSI data. Moreover, they also overlook the nonlinear correlations between bands, which can negatively impact the classification performance of the model and result in poor generalization ability.

Given the aforementioned intractable obstacles, namely, insufficient extraction of nonlinear spatial–spectral joint information, which leads to poor classification performance; a bias in the tradeoff between central and neighboring pixels, resulting in fuzzy classification of boundary pixels; and excessive spatial–spectral redundancy in HSI data, which leads to poor model generalization ability, we propose a nonlinear weighted graph convolution network based on manifold geometric regularization (MGRNWGCN) for HSIC.

Unlike other graph convolution methods for HSIC that only consider node features, the nonlinear weighted graph convolution in the MGR-NWGCN model simultaneously accounts for the relationships between nodes and edges. The MGR-NWGCN model employs cosine similarity-based nonlinear weighted graph convolution network (NWGCN) and a graph-in-graph structure to capture spectral and spatial nonlinear information, respectively. To evaluate the ability of the NWGCN module to extract nonlinear information during

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the training phase and constrain the model to shape a more accurate similarity-measuring manifold surface, a manifold geometric regularization (MGR) module is proposed. The MGR module has two submodules: manifold curvature regularization (C-MGR) and manifold intrinsic dimensionality regularization (ID-MGR). The C-MGR module, as one of the proposed MGR submodules, regularizes the curvature of the feature manifold surface generated by the model, thereby constraining the model to learn a flatter feature representation manifold. The ID-MGR module, as another proposed MGR submodules, regularizes the intrinsic dimension (ID) of the feature manifold surface generated by the model, embedding noise onto the surface of the feature manifold.

The main contributions of this article are summarized as follows.

1) To fully extract the nonlinear spatial–spectral information of hyperspectral data, an NWGCN module is designed, which is based on a graph-in-graph structure. The NWGCN performs weighted edge normalization to trade off between central superpixel information and neighboring information using cosine similarity. Therefore, the nonlinearity of spectral and spatial features is preserved, thereby enhancing classification performance.
2) To improve the accuracy of edge similarity measurement while enhancing the generalization capability of the model, the C-MGR module is implemented by introducing curvature metric of manifold in high-dimensional non-Euclidean space specially for HSI data, which estimates the curvature of the global features manifold surface by assessing the local manifold curvature of the training samples features. By minimizing the geometric regularization term based on local curvature, the model is constrained to shape a more flattened manifold surface, thereby improving the accuracy of similarity measurement for non-Euclidean data.
3) To improve the accuracy of edge similarity measurement and eliminate redundant information, the ID-MGR is developed by introducing an ID metric of a low-dimensional manifold surface in high-dimensional feature space. By minimizing the geometric regularization term based on ID, which is expanded due to the noise distribution, and embedding noise samples onto the low-dimensional manifold surface, the sensitivity of the model to noise interference is mitigated.

The remainder of this article is organized as follows. In Section II, the GCN-related knowledge and geometric metric methods on manifolds are introduced in detail. In Section III, the proposed MGR-NWGCN method is presented in detail. In Section IV, extensive experiments are reported and analyzed. In Section V, the conclusions are drawn.

II. Related Work

In this section, the basic definitions and the preliminaries of GCN, along with manifold geometric estimation methods for neural networks, are introduced.

A. Graph Convolutional Network

The data, which can be described as an undirected graph structure, are defined as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$. The $\mathcal{V}$ denotes the N nodes in the graph, and the $\mathcal{E}$ denotes the edges between nodes. The adjacency matrix $\mathbf{A}$ is composed of Boolean type or weighted data, and the size of the matrix is $\mathrm{N} \times \mathrm{N}$. Methods for constructing adjacency matrix include spatial adjacency (SA)-based and k-nearest neighbor (k-NN)-based. The SA-based method constructs the adjacency matrix based on spatial proximity, typically linking neighboring superpixels within a fixed local window to preserve spatial structure [45], [46]. The k-NN-based method constructs the graph by connecting each node to its k-NN in the feature space, capturing semantic similarity between nodes [47], [48]. The degree matrix is defined as $\mathbf{D}$, a specific type of diagonal matrix that represents the degree of each node in a graph, meaning the number of edges that connect the node to other nodes in the graph.

The feature matrix of nodes in the graph is defined as $\mathbf{H} \in \mathbb{R}^{(N \times F)}$, where $\mathbf{F}$ is the number of features per node.

The core idea of graph neural networks (GNNs) is to aggregate the features of each node with its neighboring nodes to form a new node representation. A GNN layer aggregates the features $\mathbf{H}^{(l)}$ to form the new node representation $\mathbf{H}^{(l+1)}$ using the following formula:

$$\mathbf{H}^{(l+1)} = f(\mathbf{A}, \mathbf{H}^{(l)}) \tag{1}$$

where f may include message passing, where each node receives messages from its neighboring nodes and aggregates these messages into a new node representation [49], [50], information aggregation [51], [52], [53], feature updating [54], [55], [56], and other specific implementation depending on the architecture and design goals of GNNs.

Graph Laplacian matrix is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$. By sampling global nodes based on the values of the adjacency matrix and integrating feature information, convolutional operations on irregular graph structures are given by

$$\mathbf{L}_{\text{sym}}\mathbf{X} = \mathbf{D}^{-\frac{1}{2}}\mathbf{L}\mathbf{D}^{-\frac{1}{2}}\mathbf{X} \tag{2}$$

where $\mathbf{L}_{\text{sym}}$ is the symmetric normalized form of the matrix $\mathbf{L}$. Equation (2) can be further optimized by using the first two terms of Chebyshev polynomials for approximation, and the convolutional operations can be simplified as follows:

$$\begin{aligned} g_\theta * \mathbf{X} &= \theta_0\mathbf{X} - \theta_1(\mathbf{L}_{\text{sym}} - \mathbf{I})\mathbf{X} \\ &= \theta_0\mathbf{X} - \theta_1(\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}})\mathbf{X} \\ &= (\mathbf{I} + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}})\mathbf{X} \end{aligned} \tag{3}$$

where θ_1 is defined as $-\theta_0$, and $(\mathbf{I} + \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)})$ is expressed as $\tilde{\boldsymbol{D}}^{-(1/2)}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-(1/2)}$. Based on this, the propagation rule for graph convolution is defined as

$$\mathbf{H}^{l+1} = \sigma\left(\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\tilde{\boldsymbol{A}}\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\boldsymbol{H}^{l}\boldsymbol{W}^{l}\right) \tag{4}$$

where $\mathbf{W}^l$ represents the learnable parameters.

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B. Manifold Curvature Estimation

Studies have already demonstrated that effective blocks shape more flattened manifold surfaces in the DL community [57], [58], [59], Additionally, many computational methods have been proposed for estimating the curvature of manifolds [60], [61], [62]. Based on our research content, this section introduces an angle-based curvature calculation method.

1) Match a Local Subspace: The $\mathcal{C} = \{x_i\}_{(i=1)}^{N}$ denotes a set of points on the manifold $\mathbb{M}$, where $x_i \in \mathbb{R}^D$. For x_i, the distance matrix is defined as $\mathbf{W}_i^T\mathbf{W}_i$, where $\mathbf{W}_i = [x_1 - x_i, \ldots, x_N - x_i]$, the nearest neighboring points $\{i_1, \ldots, i_{N_i}\}$ are determined by calculating diagonal values of $\mathbf{W}_i^T\mathbf{W}_i$, and the mean of the near N_i neighboring points is μ_i. Let $\mathbf{M}_i = \left[x_{i_1} - \mu_i, \ldots, x_{i_{N_i}} - \mu_i\right]$, and the process of singular value decomposition (SVD) of $\mathbf{M}_i$ is defined as

$$\mathbf{M}_i = U_i \sum_i V_i^T \tag{5}$$

where $U_i = \left[u_1, \ldots, u_{N_i}\right]$. Then, the local subspace can be matched as $\mathcal{S}_i = \text{span}\{u_1, \ldots, u_{d_i}\}$ and $\mathcal{S}_j = \text{span}\{u_1, \ldots, u_{d_j}\}$, where d_i and d_j are the dimension numbers of the two local subspaces.

2) Weighted Angle: $\mathbf{M}_i$ and $\mathbf{M}_j$ are the matrices of data points in the subspaces $\mathcal{S}_i$ and $\mathcal{S}_j$. Based on (5), $\mathbf{M}_i = U_i \sum_i V_i^T$ and $\mathbf{M}_j = U_j \sum_j V_j^T$. Let $Q = \left(U_i \sum_i\right)^T \left(U_j \sum_j\right)$, and the SVD of Q is $Q = U_Q \sum_Q V_Q^T$. Therefore, the weighted angle between $\mathcal{S}_i$ and $\mathcal{S}_j$ can be expressed as

$$\theta_{ij} = \arccos\left(\frac{\text{tr}\left(\sum_Q\right)}{\text{tr}\left(\sum_i^T \sum_j\right)}\right) \tag{6}$$

where tr$(\cdot)$ is defined as the sum of the elements on its main diagonal. The weighted angle is given by

$$\theta_i = \frac{\sum_{j=1}^{i_{N_i}} \theta_{ij}}{N_i} \tag{7}$$

where tr denotes the trace of the matrix. The curvature of $\mathbb{M}$ is defined as

$$r_{\mathbb{M}} = \frac{\sum_{i=1}^{N}(\theta_i)}{N}. \tag{8}$$

C. Manifold ID Estimation

The HSI data are embedded into a feature space $\mathbb{R}^d$ to extract feature vectors using DL-based methods, and the vectors with low dimension demonstrate improved classification performance. The ID is defined as $p(p < d)$, representing the minimum number of free variables or parameters needed to describe the data while preserving its essential structure or variability. The following focuses on introducing related algorithms for the estimation of ID.

The methods based on projection techniques can project data to the best subspace by minimizing the information loss [63]. The maximum likelihood estimation (MLE) algorithm employs input data to estimate parameter values that maximize the probability of the observed data [64]. The MLE estimator of a sample x is

$$\widehat{\text{ID}}_{\text{MLE}}(x) = -\left(\frac{1}{k}\sum_{k=1}^{k} \ln\frac{r_i(x)}{r_k(x)}\right)^{-1} \tag{9}$$

where $r_i(x)$ denotes ith nearest neighbor distance of x and $r_k(x)$ is the maximum neighbor distance of x. However, the MTE method is only applicable when the neighborhood size contains hundreds of sample points and is not suitable for small-scale sample neighborhoods. Therefore, Amsaleg et al. [65] proposed a extreme-value-based local ID (LID) model

$$\widehat{\text{ID}}_{\text{LID}}(x') = -\left(\frac{1}{|V_*|^2}\sum_{\substack{v,w\in V_* \\ v\neq w}}\left[\ln\frac{d_{x',r}(v,w)}{r_k(x')} + \ln\frac{d_{x',r}(2x'-v,w)}{r_k(x')}\right]\right)^{-1} \tag{10}$$

where V denotes the neighborhood samples, x' is the neighborhood center, and $V_* = V \bigcup \{x'\}$, $d_{x',r}(v,w)$ is defined as

$$d_{x',r}(v,w) = \frac{r_k(x')(w-v)\cdot(w-v)}{2(x'-v)\cdot(w-v)}. \tag{11}$$

III. Methodology

In this section, the proposed MGR-NWGCN method is introduced in detail, which includes four modules called NWGCN, MGR, C-MGR, and ID-MGR.

As can be seen in Fig. 2, an HSI is first segmented into a graph-structured data consisting of nodes and edges based on the simple linear iterative clustering (SLIC) algorithm. A node represents a superpixel, and the pixels within a superpixel share features generated by internal graph convolution. Integrating abstract features from internal and external graph convolutions facilitates the extraction of nonlinear spatial and spectral features. The external graph convolution, referred to as NWGCN, is based on normalizing edge-based cosine similarity weights to integrate information from neighboring superpixels.

After extracting the nonlinear spatial and spectral features, these features are distributed on a low-dimensional manifold embedded in a high-dimensional space. The geometric properties of the manifolds, including curvature and ID, are then measured to constrain the model learning and enhance feature representation. MGR strategy in the proposed method is implemented using two modules, namely, C-MGR and ID-MGR. First, the C-MGR module regularizes the curvature of the feature manifold surface, constraining the model to learn a flatter representation. This enables nonlinear spatial–spectral features to be measured linearly and facilitates convolutional operations to more accurately integrate local features, thereby enhancing the classification performance of the model. Then, the ID-MGR module is developed to regularize the ID of the feature manifold generated by the model. By embedding noise onto the surface of the feature manifold, it eliminates redundant information that hinders HSIC.

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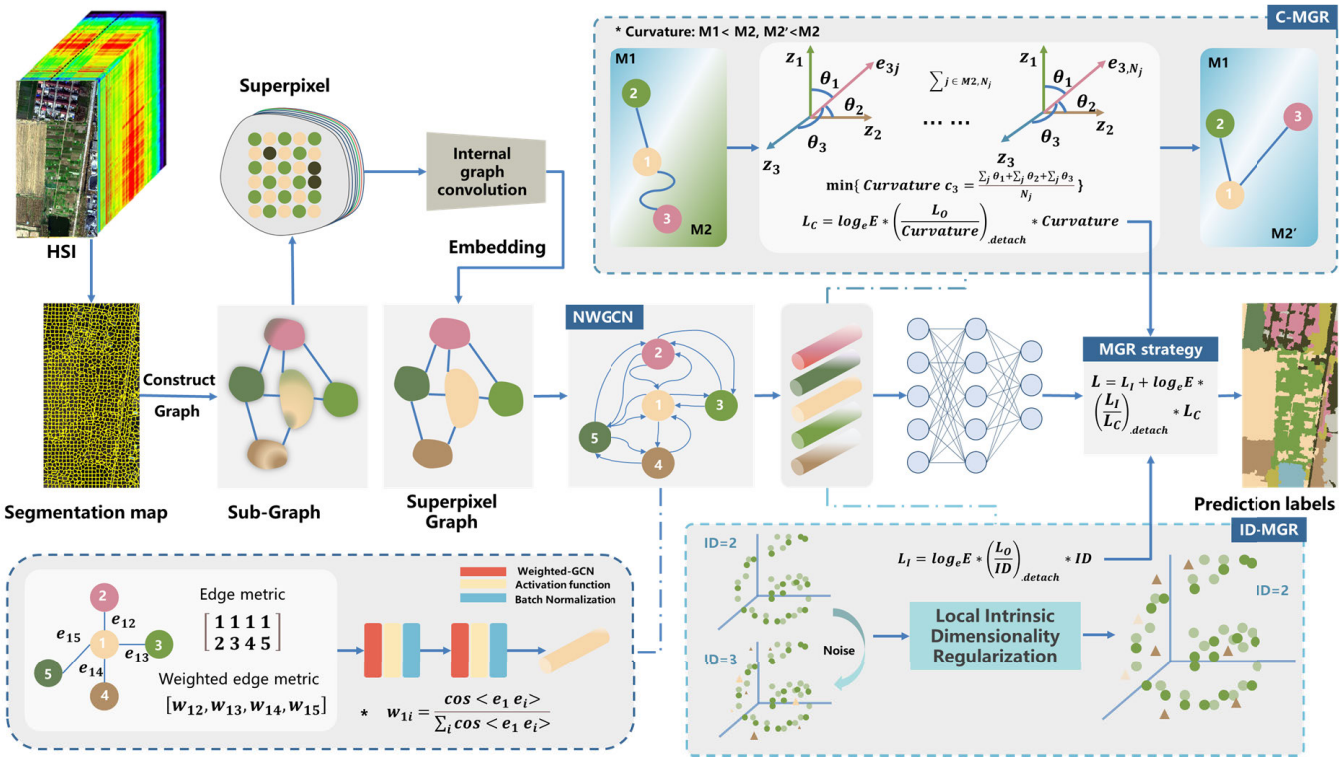


Fig. 2. Overall framework of the MGR-NWGCN method. The framework consists of four modules: NWGCN, C-MGR, ID-MGR, and manifold geometry regularization strategy (MGR). The NWGCN module can extract nonlinear spatial–spectral joint features, and features are distributed on low-dimensional manifolds in high-dimensional space. The C-MGR and ID-MGR modules measure the curvature and ID of the generated feature manifolds, respectively, and incorporate them as constraint terms to regulate model training in the subsequent MGR strategy.

A. Definitions of Terms

An HSI data is a 3-D cube with dimensions of $\mathbf{H} \times \mathbf{W} \times \mathbf{C}$, where $\mathbf{H}$ is the height, $\mathbf{W}$ is the width, and $\mathbf{C}$ is the number of bands, defined as $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{(\mathbf{H}\times\mathbf{W})}\} \in \mathbb{R}^{(\mathbf{H}\times\mathbf{W})\times\mathbf{C}}$. After the process of superpixel segmention, the $\mathbf{W} \times \mathbf{H}$ pixels in HSI change to $\mathbf{S} = \{\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_n\} \in \mathbb{R}^{n\times h}$, which has n superpixels, and h is the number of hidden layers in the internal graph convolution. $\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n\} \in \mathbb{R}^{n\times h}$ is the output of the proposed NWGCN module.

B. Nonlinear Weighted Graph Convolutional Network

Based on the input HSI data $\mathbf{X}$, an external feature matrix $\mathbf{S}$ is constructed as follows:

$$\mathbf{S} = \text{pool}_{\text{mean}}\left[(\tilde{D}_{\text{in}}^{-\frac{\mathbf{1}}{\mathbf{2}}}\tilde{A}_{\text{in}}\tilde{D}_{\text{in}}^{-\frac{\mathbf{1}}{\mathbf{2}}})\mathbf{X}\mathbf{W}_{\text{in}}\right] \tag{12}$$

where $\tilde{\boldsymbol{D}}_{\text{in}}$ is the degree metric of the internal graph convolution, $\tilde{\boldsymbol{A}}_{\text{in}}$ is the adjacency matrix, and $\mathbf{W}_{\text{in}}$ is the learnable parameter. Through (12), homogeneous pixels within a superpixel share the same features, adjacency relationships are established between superpixels, and the model can capture the nonlinear spatial relationships.

Based on $\mathbf{S}$, we iteratively select the edge pixels of each superpixels and construct edge connection $E \in \mathbb{R}^{2\times k}$ in eight directions, where $k \in \mathbb{Z}, 1 \le k \le 8$ is the number of edges. The eight directions are derived from a 3×3 fixed neighborhood window, excluding the central pixel. Subsequently, assign each edge $e_{ij}, i, j \in [1, n]$ a normalized weight coefficient based on cosine similarity

$$a_{ij} = \begin{cases} \dfrac{\text{cons} < \mathbf{s_i}, \mathbf{s_j} >}{\sum_{j=0} \cos < \mathbf{s_i}, \mathbf{s_j} >}, & \sum_{j=0} \cos < \mathbf{s_i}, \mathbf{s_j} > \neq 0 \\ 1, & \sum_{j=0} \cos < \mathbf{s_i}, \mathbf{s_j} > = 0 \end{cases} \tag{13}$$

where a_{ij} is an element of weight coefficient matrix $\tilde{\boldsymbol{A}}_{\text{ex}} \in \mathbb{R}^{n\times n}$, which is a sparse matrix. It is worth noting that $a_{ij} \neq a_{ji}$, and matrix $\tilde{\boldsymbol{A}}_{\text{ex}}$ cannot be expressed as the form of a real symmetric matrix. Feature aggregation of neighboring nodes can be achieved through $\tilde{\boldsymbol{A}}_{\text{ex}}\mathbf{S}$

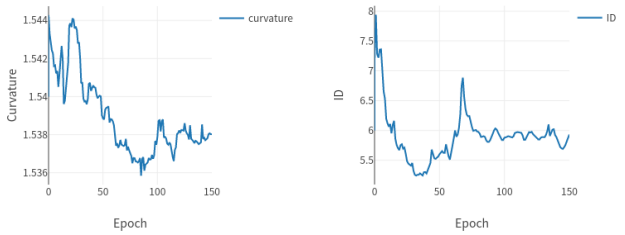
$$\tilde{\boldsymbol{A}}_{\text{ex}}\mathbf{S} = \begin{bmatrix} \sum_{(1,j\in[1,n])}(\mathbf{s}_1 + a_{1j}\mathbf{s}_j) \\ \sum_{(2,j\in[1,n])}(\mathbf{s}_2 + a_{2j}\mathbf{s}_j) \\ \cdots \\ \sum_{(i,j\in[1,n])}(\mathbf{s}_i + a_{\text{ij}}\mathbf{s}_j) \\ \cdots \\ \sum_{(n,j\in[1,n])}(\mathbf{s}_n + a_{nj}\mathbf{s}_j) \end{bmatrix}. \tag{14}$$

Then, one layer in the external graph convolutional network can be expressed as

$$\mathbf{Z}^{(l+1)} = BN(\ \text{relu}(\ \tilde{\boldsymbol{A}}_{\text{ex}}^{l}\mathbf{Z}^{l}\mathbf{W}_{\text{ex}}^{l}\ )) \tag{15}$$

where $\mathbf{W}_{\text{ex}}^{l}$ is a trainable matrix, $\mathbf{Z}^{l}$ is a feature matrix of the lth layer number, and $\mathbf{Z}^{(l+1)}$ is a feature matrix of the $(l+1)$th layer number, where l is set to 2, which can capture long-distance features without degrading model performance. The

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(a) (b)

Fig. 3. (a) Variation of curvature with training epoch. (b) Variation of ID with training epoch.

normalization of weighted edges $\tilde{\boldsymbol{A}}$ in (15) is proposed, which is noteworthy that matrix $\tilde{\boldsymbol{A}}$ is being measured dynamically based on feature MGR strategy in high-dimensional space, and the detailed information of the MGR strategy will be introduced in Section III-C. Based on this, the module can capture the nonlinear spectral joint features of neighboring superpixels.

In summary, the proposed NWGCN module can obtain nonlinear spatial information through internal and external convolutions of the superpixels and extract nonlinear spectral information by dynamically weighting and normalizing the adjacency matrix.

C. Manifold Geometric Regulation

As aforementioned, given a graph structure that includes superpixel data $\mathbf{S}$ and an adjacency weight matrix $\tilde{\boldsymbol{A}}$, a trained model $logits = \{g(\mathbf{s}, \theta_1), c(\mathbf{z}, \theta_2)\}$ is generated through MGR strategy constraints, where $g(\mathbf{s}, \theta_1)$ denotes the trained NWGCN model, the output of the network g is $\mathbf{Z}$, and $c(\mathbf{z}, \theta_2)$ denotes the classifier of the proposed MGR-NWGCN model.

To constrain the model to reduce the curvature of the manifold surface $\mathbb{M}^d$ on which the training feature samples $\mathbf{Z}_t$ reside, while simultaneously compressing noise in order to decrease the intrinsic dimensionality of the manifold surface $\mathbb{M}^d$, the MGR strategy is proposed. The pseudocode for the MGR strategy is shown in Algorithm 1. First, the L_1 penalty is added to drop out nontight connection of the internal graph convolution

$$L_1 = L_{\text{org}} + \lambda \sum_{i,j} \left| \mathbf{W}_{\text{in}ij} \right| \tag{16}$$

where L_{org} is the original cross-entropy of classification loss. Then, the strategy of the manifold geometric regulation is given by

$$L_{\text{ID}}^l = L_1 + \frac{\log_{\tau_2} l}{\widehat{\text{ID}}_t^l \mid L_1} \times \widehat{\text{ID}}_t^l \tag{17}$$

$$L^l = L_{\text{ID}}^l + \frac{\log_{\tau_1} l}{C_t^l \mid L_{\text{ID}}^l} \times C_t^l \tag{18}$$

where τ_2 and τ_1 are the hyperparameters for ID-MGR and C-MGR specifically. The variation of the curvature of the manifold with training epochs is shown in Fig. 3(a), and the change in ID over training epochs is illustrated in Fig. 3(b). Both of them exhibit nonlinear variations, which result from the balance between the classification loss and the MGR term.

Algorithm 1 Pseudocode for the MGR Strategy

Input: Data $X \in \mathbb{R}^{N \times D}$ is the lth features extracted by NWGCN module, neighbors N_r, coefficient τ_1, coefficient τ_2, original loss $L_{original}$.
Output: lth Loss L^l.

```
if epoch > e_2 then
    Compute ι ←pmg.ID_TLE(X);
    L_ι^l = L_original + log_{τ_2} l / (ι|L_original) × ι
else L_ι^l = L_original
end if
if epoch > e_1 then
    Fit nearest N_r neighbors model to X
    for i ∈ [1, N] do
        Compute neighbors M_i of point i
        Perform SVD: U_i, Σ_i = SVD(M_i)
        for neighbor j ∈ M_i do;
            Compute neighbors M_j and SVD: U_j, Σ_j
            Compute Q = U_i^⊤ U_j and SVD of Q
            Compute angle θ using Σ_Q
            Accumulate κ_i ← κ_i + θ
        end for
        Curvature κ_i ← κ_i / k
    end for
    Compute κ = mean(κ_i)
    L^l = L_ι^l + log_{τ_1} l / (κ^l |L_ι^l) × κ^l.
else L = L_ι^l
end if
```

D. Manifold Curvature Regularization

This section introduces the C-MGR module, which results in a flat feature representation manifold for $\mathbf{Z}$.

The flat feature representation manifold means that the features on the original surface are projected onto a surface with more accurate measurements. The specific implementation details are illustrated in Fig. 4. Let $\mathbf{Z}_t = \{\mathbf{z}_i\}_{i=1}^{N_t}$, where $\mathbf{Z}_t$ is the N_t samples on a manifold $\mathbb{M}^d$ used to train the θ_1 of the $g(\mathbf{s}, \theta_1)$ network, $\mathbf{Z}_t^i = \{\mathbf{z}_j\}_{j=1}^{N_r}$ is the neighborhood samples of $\mathbf{z}_i$, and the curvature of $\mathbb{M}^d$ in the lth epoch of the training stage is defined as follows:

$$C_t^l = \sum_{i=1}^{N_t} \sum_{j=1}^{N_r} \frac{\arccos\left(\frac{\text{tr}(\sum_Q)}{\text{tr}(\sum_i^T \sum_j)}\right)}{N_r N_t} \tag{19}$$

where N_r is the number of the neighborhood samples taken for $\mathbf{z}_i$, and the principle for selecting N_r will be given in Section IV-C2. The SVD is a fundamental matrix factorization technique. By using SVD, let $Q = (U_i \sum_i)^T (U_j \sum_j)$, U_i is an $N_r \times N_r$ unitary matrix of subspace $\mathbb{M}_i = \{z_i - \mu_i\}_i^{N_r}$, its columns are orthogonal and of unit norm, and $\sum_i$ is an rectangular diagonal matrix with nonnegative values along its diagonal, which are arranged in descending order. U_j and $\sum_j$ represent the unitary matrix and rectangular diagonal matrix of $\mathbf{Z}_t^i$, respectively. Therefore, Q is defined as $Q = U_Q \sum_Q U_Q^T$ with its SVD.

To achieve a flatter feature manifold representation in model shaping, the C-MGR module is proposed. It constrains the

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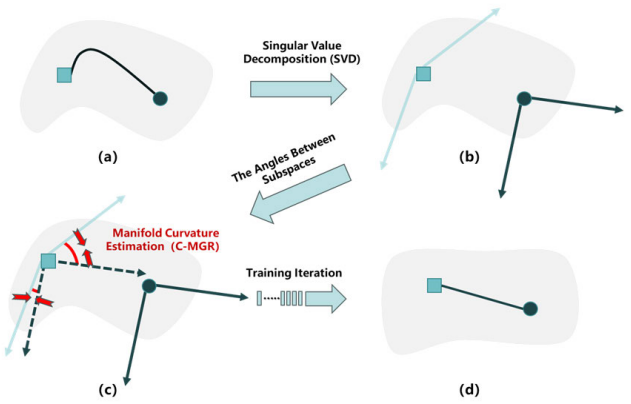


Fig. 4. Illustration of the C-MGR module. (a) Two superpixel features on the manifold surface. (b) Two subspaces are spanned by the adjacent feature sets of the two superpixels. (c) Constrain the angle between the two feature subspaces to be smaller. (d) Obtain a flatter feature manifold.

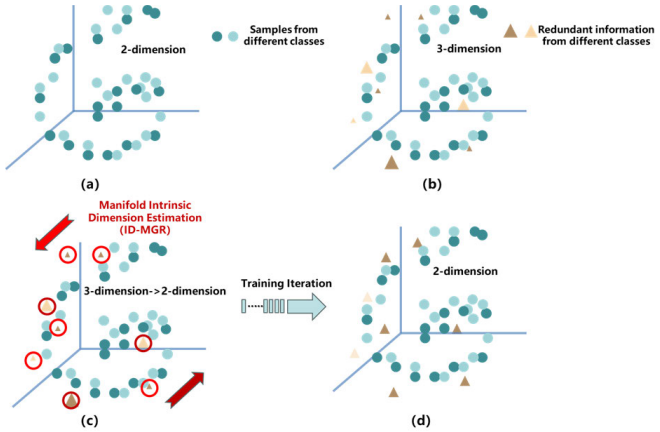


Fig. 5. Illustration of the ID-MGR module. (a) Feature manifold with an ID of 2. (b) Interference of redundant information increases the ID of the manifold to 3. (c) Constraint is imposed to project redundant information onto a 2-D manifold surface, thereby limiting its impact. (d) Generated 2-D feature manifold.

model to reduce the curvature of the feature manifold by measuring the manifold curvature at the lth iteration as a regularization term, which is given by

$$L^l = L_1^l + \frac{\log_{\tau_1} l}{C_t^l \mid L_1} \times C_t^l. \tag{20}$$

The L_1 is the original loss function based on $c(\mathbf{z}, \theta_2)$. The $C_t^l \mid L_1$ means that C_t^l divides L_1 without leaving a remainder, aiming to keep them within the same order of magnitude. In the later stages of model training, the training loss will decrease to a lower order of magnitude, and the manifold curvature regularization term needs to align with it to prevent a sudden deterioration of the performance.

E. Manifold ID Regularization

On the surface of the feature manifold $\mathbb{M}^d$, besides the feature samples $\mathbf{Z}_t$, there still exist noise with complex distribution patterns. To eliminate redundant information during the representation process of the training model, in this section, the ID-MGR module is introduced, and a regularization term $\widehat{\mathrm{ID}}$ is proposed to constrain the model. As shown in Fig. 5, it is based on the ID of the manifold surfaces $\mathbb{M}^d$ and ensures the generation of manifold surfaces with low ID.

Let $\mathbf{Z}_t^n = \{\mathbf{z}_k\} \subseteq \mathbf{Z}_t$ denote a set of noises on the manifold $\mathbb{M}^d$, and the ID of the $\mathbf{z}_i$ is defined as

$$\widehat{\mathrm{ID}}_{\mathbb{M}^d}(\mathbf{z}_i) = -\left(\frac{1}{|\mathbf{Z}_t|^2} \sum_{\substack{\mathbf{z}_j \in \mathbf{Z}_t \setminus \mathbf{Z}_t^n, \\ \mathbf{z}_r \in \mathbf{Z}_t \\ \mathbf{z}_j \neq \mathbf{z}_r}} \left[\ln \frac{d_{\mathbf{z}_i, r}(\mathbf{z}_j, \mathbf{z}_r)}{r_k(\mathbf{z}_i)} + \ln \frac{d_{\mathbf{z}_i, r}(2\mathbf{z}_i - \mathbf{z}_j, \mathbf{z}_r)}{r_k(\mathbf{z}_i)}\right]\right)^{-1} \tag{21}$$

where $\mathbf{z}_j \in \mathbf{Z}_t$ is located within a quiet small neighborhood of $\mathbf{z}_i$. Generally, this small neighborhood does not include noise $\mathbf{z}_k$, and r is defined as the radius of the neighbor circle includes $\mathbf{z}_r$. Then, the global ID of the $\mathbf{Z}_t$ is given by

$$\widehat{\mathrm{ID}}_t^l = \frac{1}{N_t} \sum_{i=1}^{N_t} \widehat{\mathrm{ID}}_{\mathbb{M}^d}(\mathbf{z}_i). \tag{22}$$

During the lth iteration of model training, a feature manifold surface is shaped, and an $\widehat{\mathrm{ID}}_t^l$ is utilized to measure the ID of the training feature samples on the surface. To constrain the model to compress noise into a low-dimensional space, thereby eliminating redundant information, the lth layer of the proposed ID-MGR is given by

$$L^l = L_1^l + \frac{\log_{\tau_2} l}{\widehat{\mathrm{ID}}_t^l \mid L_1} \times \widehat{\mathrm{ID}}_t^l. \tag{23}$$

IV. Experiments

In this section, the proposed MGR-NWGCN model is compared with nine methods on four datasets. First, Pavia University (PaviaU), Xuzhou, and Kansas datasets are introduced, followed by a description of the experimental settings and parameter selection. Then, the performance of the four modules is evaluated through module ablation. Finally, the comparative results and their analysis are discussed in Section IV-E on comparative experiments.

A. Dataset Description

1) Pavia University: The PaviaU dataset [66] was collected by German Airborne Reflective Optics Spectrographic Imaging System (ROSIS-03) during imaging of the city of Pavia, Italy, in 2003. The spectral imager captured continuous imaging across 115 bands within the wavelength range of 0.43–0.86 μm, with a spatial resolution of 1.3 m. The PaviaU dataset image consists of 610 $\times$ 340 pixels with 115 spectral bands (although 12 bands are typically excluded due to noise, leaving 103 bands for use).

2) Xuzhou: The Xuzhou dataset [67], [68][1] was gathered by an airborneHySpex SWIR-384 and HySpex VNIR-1600 imaging spectrometer over the peri-urban area of Xuzhou in November 2014. The image obtained by HySpex VNIR-1600 with 0.19-m spatial resolution has 160 bands, which cover the spectral range of 415–992 nm, and the image

[1]https://ieee-dataport.org/documents/xuzhou-hyspex-dataset#files

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR PAVIAU DATASET

No.	Color	Land cover type	Train	Test
1		Asphalt	10	6621
2		Meadows	10	18639
3		Gravel	10	2089
4		Trees	10	3054
5		Painted metal sheets	10	1335
6		Bare Soil	10	5019
7		Bitumen	10	1320
8		Self-Blocking Bricks	10	3672
9		Shadows	10	937
Total			90	42686

obtained by HySpex SWIR-384 with 0.73-m spatial resolution has 288 bands, which cover the spectral range of 952–2508 nm. After the preprocess and resample, the two images were merged into one image. This final image comprises 500 × 260 pixels, boasting an exceptionally high spatial resolution of 0.73 m/pixel. After wiping off the spectral overlap region spanning from 415 to 2508 nm, the experiment utilized 436 spectral bands.

3) Kansas: The Kansas AHSI dataset [69] was acquired by the visible–shortwave infrared (SWIR) AHSI instrument, designed by Shanghai Institute of Technical Physics, Chinese Academy of Sciences. This instrument served as a key payload on the GF-5 satellite. The dataset consists of 340 × 650 pixels and covers a spectral range of 400–2500 nm, encompassing 330 spectral bands to characterize the solar reflective regime. The spectral resolution is approximately 5 nm for the visible and near infrared (VNIR) region (400–1000 nm) and 10 nm for the visible–SWIR region (1000–2500 nm).

4) HC: The Holden Crater (HC) dataset [70] was acquired by the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) aboard the Mars Reconnaissance Orbiter (MRO) with a high spatial resolution of 18 m/pixel. The dataset includes 440 bands ranging from 0.36 to 3.93 μm and consists of 418 × 595 pixels. The dataset focuses on the Holden Crater region, a geologically significant area known for its sedimentary deposits and ancient lakebed features, which provide insights into Mars' hydrological history.

The pseudocolor images and ground-truth images of each dataset are shown in Figs. 6–9, respectively. The labeled samples in PaviaU, Kansas, and HC datasets are relatively scattered, while those in Xuzhou dataset are more continuous.

B. Experimental Settings

The land-cover categories of the datasets, along with the number of samples used for training and testing for each category, are shown in Tables I–IV, respectively. For each category in each dataset, only ten samples are used for training to assess the model's performance under the condition of limited labeled samples. To quantitatively analyze the performance of the trained model, we introduce overall accuracy (OA) [71], average accuracy (AA) [71], and Kappa coefficient (Kappa) [72] as evaluation metrics. The OA indicates the ratio of correctly classified samples to the total test samples, defined as

$$\mathrm{OA} = \frac{\text{Number of Correct Predictions}}{\text{Total Test Samples}} \times 100\%.a. \quad (24)$$



(a) (b)

Fig. 6. (a) Pseudocolor image. (b) Ground-truth map of PaviaU dataset.

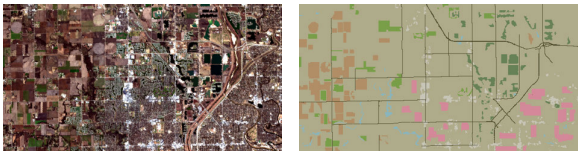
TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR XUZHOU DATASET

No.	Color	Land cover type	Train	Test
1		Bareland-1	10	26386
2		Bareland-2	10	6980
3		crops-1	10	13174
4		crops-2	10	4767
5		Red-tiles	10	3060
6		Trees	10	2426
7		Coals	10	2773
8		Cement	10	5204
9		Lakes	10	4017
Total			90	68787



(a) (b)

Fig. 7. (a) Pseudocolor image. (b) Ground-truth map of Xuzhou dataset.



(a) (b)

Fig. 8. (a) Pseudocolor image. (b) Ground-truth map of Kansas dataset.

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TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR KANSAS DATASET

No.	Color	Land cover type	Train	Test
1		Crop	10	5193
2		Meadow	10	5458
3		Business Area	10	4685
4		Bare Soil	10	12078
5		Residential Area	10	8225
6		Water	10	1673
7		Road	10	4571
Total			70	41883

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR HC DATASET

No.	Color	Land cover type	Train	Test
1		Analcime	10	930
2		Plagioclase	10	1462
3		Prehnite	10	1550
4		High-CaPyroxene	10	6953
5		Serpentine	10	8687
6		Margarite	10	448
Total			60	20030

The AA indicates the mean of classification accuracies across all classes, calculated as

$$\mathrm{AA} = \frac{1}{C}\sum_{i=1}^{C}\frac{\text{Correct Predictions in Class } i}{\text{Test Samples in Class } i} \times 100\%. \tag{25}$$

The Kappa is a statistical measure of interrater agreement that accounts for chance, computed by

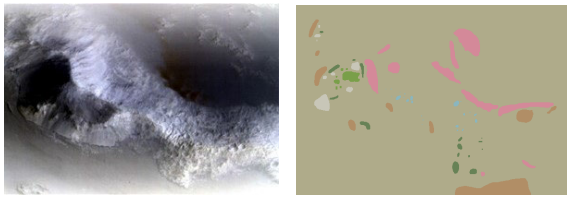
$$\mathrm{Kappa} = \frac{\mathrm{OA} - p_e}{1 - p_e} \tag{26}$$

where p_e is the hypothetical probability of random agreement.

A detailed overview of the hardware and software environments is provided as follows, as well as the relevant configuration information used in the experiments. The experiments were run on a server with an NVIDIA GeForce RTX-3090 GPU and 56-GB memory. The operating system is Ubuntu 20.04 LTS, CUDA 11.1 is adopted to leverage the computing capabilities of the GPU, PyTorch 1.10.1 is chosen as the DL framework, and the Python version is 3.9.12. The experiments also relied on Python libraries such as NumPy, SciPy, Matplotlib, and Scikit-learn.

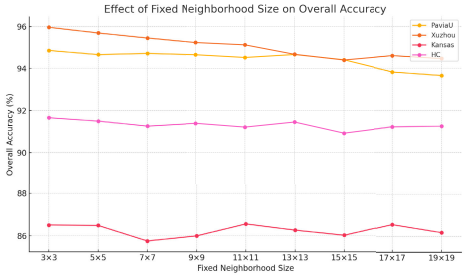
C. Parameter Analysis

1) Fixed Neighborhood Size of Adjacency Matrix Construction: A larger fixed window allows the model to capture information from more distant connections within a single graph convolution. However, this inevitably introduces irrelevant information, which may interfere with the model's ability to effectively aggregate local features. The size of the fixed neighborhood is also closely related to both the computational complexity of the model and its feature aggregation capability. For a single graph convolutional layer, the computational complexity is $\mathcal{O}(Nc^2)$, where N denotes the number of superpixels and c represents the size of the fixed adjacency window. Therefore, a larger window size leads to a proportional increase in computational complexity. Based on the experiments in Fig. 10, c is set to 3 in this work. Using an overly large neighborhood may lead to a decline in classification performance. While larger neighborhoods result

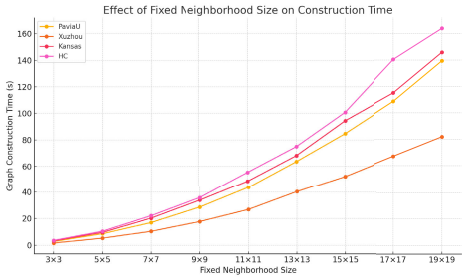


(a) (b)

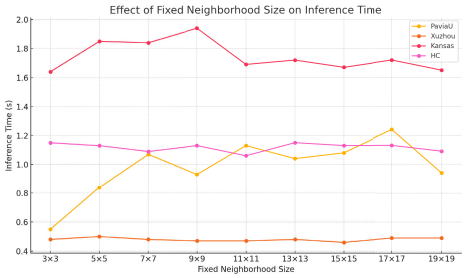
Fig. 9. (a) Pseudocolor image. (b) Ground-truth map of HC dataset.



(a)



(b)



(c)

Fig. 10. Tradeoff between computational efficiency and feature integration under different fixed neighborhood sizes. (a) OA. (b) Graph construction time. (c) Inference time, evaluated on four datasets.

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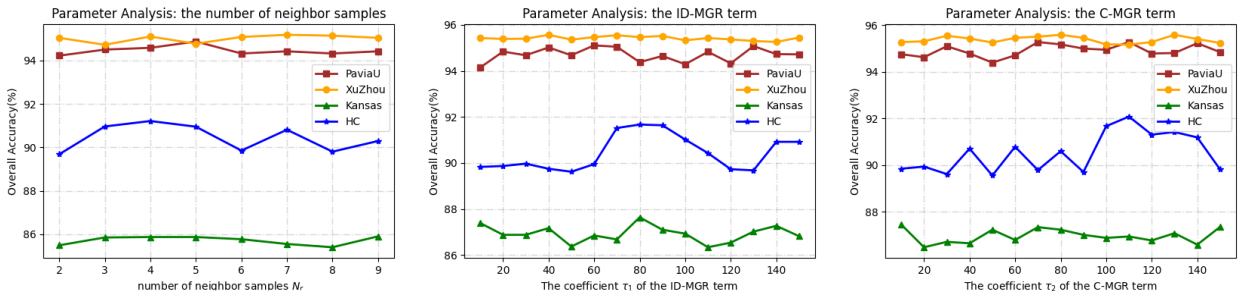


Fig. 11. Classification performance of the proposed MGR-NWGCN method with (a) different number of neighbor samples N_r, (b) different coefficient τ_1 of the ID regulation term, and (c) different coefficient τ_2 of the curvature regulation term on the four datasets. All axes are dimensionless unless specifically noted.

in longer graph construction times [48], their effect on model inference time remains minimal.

2) Number of Neighbor Samples in the Local Manifold: The curvature estimation, which is adopted for manifold ensemble regularization, requires selecting a local nearest neighborhood for calculation. A larger N_r (the number of neighboring samples) in the neighborhood will increase the computational complexity, while a smaller N_r may affect the accuracy of the estimation. Since there are ten training samples for each category, the selection of parameter N_r is constrained to the integer sequence within the range of [2, 9]. To evaluate the impact of N_r on classification performance, systematic experiments were conducted across four benchmark datasets. As illustrated in Fig. 11(a), when N_r within the range [2, 5], the OA stabilizes and approaches higher accuracy. Therefore, parameter N_r is set to 4. The principle for selecting the parameter was to choose the smallest possible number of neighboring samples while ensuring the accuracy of curvature estimation.

3) Parameter Selection for MGR: The C-MGR term, which is given by (18), involves four parameters, namely, e_1, e_2, τ_1, and τ_2, These parameters are used to balance the constraints among the curvature regularization term, the ID regularization term, and the original loss. Specifically, e_1 and e_2 control the starting epochs at which the manifold curvature regularization term begins to impose constraints on the model training, respectively. Based on the experiments in [73], e_1 and e_2 are set to 5 in this work. The reason for setting these two parameters is that during the initial epochs of model training, the generated manifold surface undergoes significant changes, and the estimated manifold geometric parameters are not yet capable of guiding the model to generate a manifold surface with better classification performance.

Meanwhile, $\log_{\tau_1} l$ and $\log_{\tau_2} l$ are used to modulate the intensity of the proposed C-MGR and ID-MGR terms as training progresses. Larger τ_1 and τ_2 values delay the influence of C-MGR and ID-MGR, while smaller values accelerate it. The impact of C-MGR and ID-MGR on model performance increases as the number of training epochs increases. Therefore, parameters should be set to adjust the degree of influence that the C-MGR and ID-MGR terms have the model at different training epochs. With different values of τ_1 and τ_2 selected on the four datasets, the classification results obtained are shown in Fig. 11(b) and (c). For PaviaU, Xuzhou, and Kansas datasets, the overall performance of the model is not significantly affected by different parameter selections. For dataset HC, when the parameters τ_1 are set within the range of $[70, 120]$ and τ_2 are set within the range of $[100, 140]$, the classification performance of the model exhibits minimal fluctuations. Therefore, τ_1 is set to 100 and τ_2 is set to 110 in this work.

D. Module Ablation

In this section, the impact of the proposed NWGCN, C-MGR, ID-MGR modules, and MGR strategy on the classification performance will be presented and analyzed. In the ablation study, the effectiveness of each module is verified by employing a step-by-step addition strategy. Starting from the NWGCN model, modules are incrementally integrated one at a time to evaluate their individual contributions to the overall performance.

1) NWGCN Test: In the ablation experiment targeting the NWGCN module, NWGCN leverages its graph-in-graph structure based on superpixel segmentation and normalized weighted graph convolution based on cosine similarity to extract the nonlinear spatial–spectral joint information from HSI data. A GCN model based on (2) is used to verify the improvement in classification performance attributed to the ability of NWGCN to extract nonlinear information.

2) C-MGR Test: For the ablation experiment targeting the C-MGR module, the C-MGR module constrains the model to reduce the curvature of the feature manifold, thereby improving the accuracy of edge distance measurement in the local neighborhood of superpixels, enabling more precise integration of local information and enhancing the feature representation capability of the MGR-NWGCN model. As shown in Table V, the overall performance of the model has been improved on all four datasets such as PaviaU, Xuzhou, Kansas, and HC. Specifically, on the Kansas dataset, the OA increase exceeded 2%, the AA increase exceeded 1%, and the kappa coefficient improvement was over 2%.

3) ID-MGR Test: Ablation experiment on the ID-MGR module of the proposed model MGR-NWGCN is conducted. The ID-MGR module eliminates redundant information by embedding noise onto the low-dimensional manifold surface. The results are shown in Table V. On PaviaU, Xuzhou, Kansas,

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TABLE V
ABLATION STUDY ON THREE DATASETS

Data Set	NWGCN	C-MGR	ID-MGR	regulation strategy	OA	AA	$K \times 100$
PaviaU	✓	×	×	×	93.97	95.08	92.07
	✓	✓	×	×	94.32	95.03	92.52
	✓	✓	✓	×	94.44	95.33	92.67
	✓	✓	✓	✓	94.68	95.56	93.00
Xuzhou	✓	×	×	×	94.89	96.57	93.59
	✓	✓	×	×	95.28	96.32	94.06
	✓	✓	✓	×	95.31	96.29	94.10
	✓	✓	✓	✓	95.96	96.56	94.89
Kansas	✓	×	×	×	84.16	80.49	80.81
	✓	✓	×	×	86.17	82.16	83.18
	✓	✓	✓	×	86.30	82.08	83.35
	✓	✓	✓	✓	86.51	82.39	83.59
HC	✓	×	×	×	91.12	89.13	87.99
	✓	✓	×	×	91.32	88.76	87.38
	✓	✓	✓	×	91.91	88.97	88.26
	✓	✓	✓	✓	92.05	89.33	88.48

and HC datasets, there were improvements in both OA and kappa coefficients. Additionally, there was an improvement in AA on the PaviaU and HC datasets. The limited enhancement of performance by the ID-MGR module on these four datasets is attributed to the fact that, without using a regularization strategy, the magnitude of the sum of two regularization terms exceeded the cross-entropy loss in the later stages of model training, thereby weakening the dominant role of the cross-entropy in loss. This phenomenon will be improved when a regularization strategy is employed.

4) Manifold Geometric Regulation: As previously mentioned, both the C-MGR and ID-MGR modules require an appropriate regularization strategy to ensure that they remain on the same order of magnitude as the cross-entropy loss at any stage of model training, while also guaranteeing their constraining effects on shaping of the feature manifold surface. As shown in Table V, after applying manifold geometry regularization, the OA improved by more than 1% on both the Xuzhou and Kansas datasets. Additionally, the kappa coefficients increased by more than 1% on the PaviaU, Xuzhou, and Kansas datasets.

E. Comparative Experiment

To evaluate the performance of the proposed approach, nine comparative methods for research are conducted, including kernel-based SVM [13], nonlinear PCA (NLPCA) [74], nonlinear parsimonious feature selection (NPFS) [75], adaptive mask sampling and manifold to the Euclidean subspace learning (AMS-M2ESL) [76], spectral–spatial graph reasoning network (SSGRN) [77], differentiated-scale restricted GCN (DSR-GCN) [78], graph-in-graph convolutional network (GiGCN) [79], end-to-end mixhop superpixel-based GCN (EMS-GCN) [80], and SpectralFormer [27].

The kernel-based SVM, NLPCA, and NPFS are traditional methods. The AMS-M2ESL method [76] based on the manifold pays attention to the heterogeneity between the central pixel and its neighboring pixels. The GCN-based methods, including SSGRN [77], DSR-GCN [78], GiGCN [79], and EMS-GCN [80], are used as comparison methods because GCN places greater emphasis on the utilization of nonlinear structural information in HSI data compared to other DL approaches. The transformer-based method SpectralFormer [27] leverages spectral attention mechanisms to capture long-range dependencies along the spectral dimension.

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR PAVIAU

class	PCA [13]	NLPCA [74]	NPFS [75]	AMS-M2ESL [76]	SSGRN [77]	DSR-GCN [78]	GIGCN[79]	EMS-GCN [80]	SpectralFormer [27]	MGR-NWGCN
1	93.28	80.19	63.96	96.25	**60.04**	81.35	89.44	96.28	62.63	88.19
2	85.27	93.22	74.68	99.37	77.08	79.45	90.40	**88.82**	72.82	96.05
3	**41.35**	**33.51**	47.25	66.77	71.71	**74.92**	97.86	94.13	48.40	97.21
4	70.48	68.17	72.07	78.09	92.79	81.07	**82.68**	97.26	67.91	**86.88**
5	98.14	**100.00**	**99.33**	**99.93**	99.92	**100.00**	99.40	99.73	**99.85**	**99.45**
6	52.50	38.76	**37.44**	**66.06**	84.88	98.01	96.52	99.62	60.65	98.35
7	42.00	51.43	90.68	93.58	90.53	99.39	**99.41**	**99.87**	85.45	99.23
8	57.53	54.95	62.55	72.50	86.47	49.81	94.65	99.31	**36.27**	97.22
9	**99.89**	99.68	91.68	94.73	**100.00**	79.62	90.98	97.72	95.94	90.18
OA(%)	73.74	66.98	67.71	86.50	78.65	80.53	91.72	93.89	66.86	**94.68**
AA(%)	71.16	68.88	71.07	85.25	84.82	82.62	93.98	**96.97**	69.99	95.56
Kappa $\times 100$	65.71	59.31	58.23	82.74	73.08	75.25	89.25	92.10	57.32	**93.00**

TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR XUZHOU

class	PCA [13]	NLPCA [74]	NPFS [75]	AMS-M2ESL [76]	SSGRN [77]	DSR-GCN [78]	GIGCN [79]	EMS-GCN [80]	SpectralFormer [27]	MGR-NWGCN
1	90.77	88.95	82.61	97.78	68.04	94.89	92.60	92.13	85.61	95.37
2	**97.74**	**90.71**	81.60	**100**	**97.63**	98.63	**99.42**	99.28	94.35	98.71
3	84.48	43.02	80.67	86.22	87.70	**100.00**	99.24	99.40	86.22	**99.40**
4	73.53	56.22	84.15	97.87	**50.50**	**71.52**	**89.90**	**84.82**	79.71	**89.88**
5	80.93	74.92	69.84	96.44	83.73	81.61	95.74	86.36	78.98	95.94
6	**31.82**	**18.28**	**56.35**	86.78	84.62	90.89	95.26	90.71	81.53	97.37
7	66.23	36.66	61.72	**84.41**	91.63	94.84	99.08	95.78	94.44	97.85
8	95.67	37.56	**93.46**	90.15	65.72	98.02	98.34	**99.67**	**96.20**	99.38
9	87.80	80.02	87.75	98.22	85.20	99.24	92.48	93.42	**74.64**	93.97
OA(%)	80.51	61.59	78.08	94.53	75.82	91.27	94.81	92.09	85.43	**95.96**
AA(%)	78.78	58.48	77.57	93.10	79.42	92.18	96.18	93.51	85.74	**96.56**
Kappa $\times 100$	75.70	53.67	72.77	93.94	70.28	89.00	93.48	89.99	81.82	**94.89**

TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR KANSAS

class	PCA [13]	NLPCA [74]	NPFS [75]	AMS-M2ESL [76]	SSGRN [77]	DSR-GCN [78]	GIGCN [79]	EMS-GCN [80]	SpectralFormer [27]	MGR-NWGCN
1	91.98	95.32	76.24	90.94	**96.39**	87.48	81.34	86.21	**98.21**	94.39
2	**99.79**	**99.04**	89.67	**99.25**	95.82	96.06	**92.89**	**96.22**	98.13	**95.45**
3	83.48	67.39	**89.73**	70.27	87.89	75.86	79.44	89.73	91.40	83.36
4	89.78	92.94	66.90	92.81	79.12	**96.47**	83.65	87.06	82.40	88.63
5	87.57	81.31	65.73	88.44	82.77	89.28	87.34	90.75	**40.13**	92.53
6	68.71	**32.68**	88.28	50.40	78.06	77.76	79.91	87.73	83.86	77.31
7	**47.25**	36.75	**62.44**	**45.52**	**42.28**	**31.94**	**51.46**	**40.19**	53.40	**55.18**
OA(%)	82.60	75.89	73.71	81.00	81.07	83.79	81.16	84.08	76.05	**86.51**
AA(%)	81.22	72.20	77.00	76.80	80.33	79.27	79.12	**82.56**	78.28	82.39
Kappa $\times 100$	78.94	71.11	68.66	77.16	77.22	80.13	77.20	80.67	71.38	**83.59**

TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR HC

class	PCA [13]	NLPCA [74]	NPFS [75]	AMS-M2ESL [76]	SSGRN [77]	DSR-GCN [78]	GIGCN [79]	EMS-GCN [80]	SpectralFormer [27]	MGR-NWGCN
1	84.54	66.62	70.35	45.03	92.90	**100.00**	86.30	99.14	91.49	82.67
2	50.89	39.01	84.48	**35.54**	80.92	64.43	76.83	**76.81**	79.41	**80.26**
3	81.90	51.91	**35.59**	56.33	94.65	**58.00**	**71.85**	84.77	**67.61**	80.37
4	98.19	**88.06**	**99.94**	**93.61**	79.17	74.14	93.63	83.55	87.24	94.43
5	**99.95**	86.34	79.90	88.29	**63.03**	91.49	91.26	**99.48**	91.64	95.00
6	**35.76**	**21.59**	85.75	31.70	**100.00**	99.55	**99.20**	98.66	**99.33**	**98.75**
OA(%)	87.25	71.93	83.76	71.02	74.60	81.47	89.48	91.12	87.52	**92.05**
AA(%)	75.21	58.92	76.67	58.42	66.37	81.27	86.55	**90.40**	86.11	89.33
Kappa $\times 100$	82.01	61.01	86.50	60.86	85.11	73.80	84.85	87.16	82.30	**88.48**

1) Quantitative Analysis: The model performance of the nine selected comparison methods and the proposed MGR-NWGCN method is shown in Tables VI–IX. In terms of overall performance, the proposed method outperforms the other nine comparison methods. The OA metric in the proposed method on the PaviaU dataset achieves an improvement of 0.79%–27.7% compared to other methods, an improvement of 1.43%–34.37% on the Xuzhou dataset, an improvement of 2.43%–12.8% on the Kansas dataset, and an improvement of 0.93%–21.03% on the HC dataset. The AA metric on the Xuzhou dataset shows an increase of 0.38%–38.08% in the proposed method compared to other methods. A 0.9–34.77 improvement is achieved in the Kappa $\times$ 100 metric on the PaviaU dataset, 0.95–41.22 in the Xuzhou dataset,

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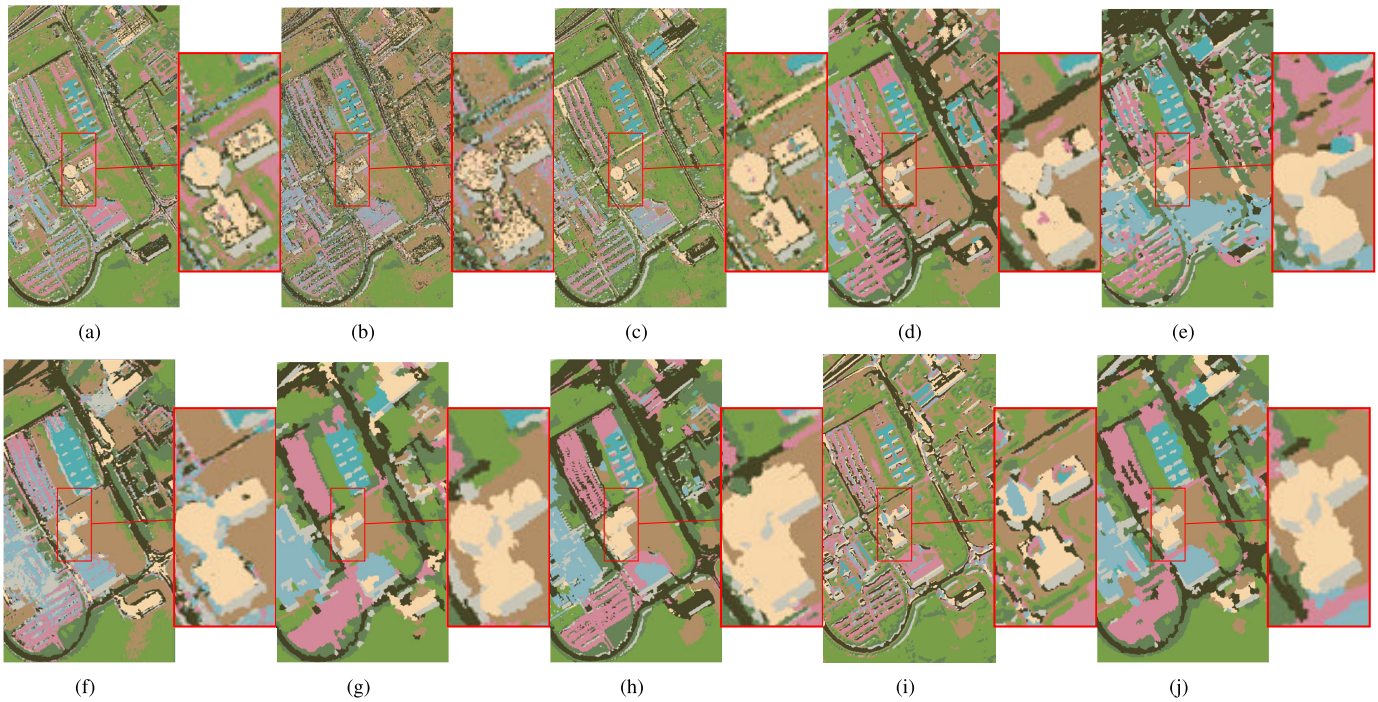


Fig. 12. Classification maps on the PaviaU dataset. (a) SVM (73.74%). (b) PCA (66.98%). (c) NPFS (67.71%). (d) AMS-M2ESL (86.50%). (e) SSGRN (78.65%). (f) DSR-GCN (80.53%). (g) GiGCN (91.72%). (h) EMS-GCN (93.89%). (i) SpectralFormer (66.86%). (j) MGR-NWGCN (94.68%).

2.92–14.93 in the Kansas dataset, and 1.32–27.62 in the HC dataset.

It should be noted that the performance on the AA metric of the proposed model for the PaviaU, Kansas, and HC datasets is not the best among all comparison methods. This is because the MGR term may lose some information during the process of flattening the manifold surface and compressing the IDs of the manifold. However, this does not lead to a significant decline in the overall performance. On the PaviaU dataset, it lags behind EMS-GCN by a difference of 1.41%, on the Kansas dataset, the difference is only 0.17%, and 1.07% on the HC dataset, while on the Xuzhou dataset, it leads by 3.05%. Moreover, as shown in the red text in Tables VI–IX, the classification accuracy gap between the best and worst categories has further narrowed compared to other comparison methods.

2) Qualitative Analysis: To better demonstrate the visualization of classification results, we present the full-map classification results of the proposed method and the comparison methods on PaviaU, Xuzhou, Kansas, and HC datasets in Figs. 12–15, and the full-map classification results include the classification results of unlabeled samples. Specifically, the proposed method exhibits clearer boundaries for local features. As shown in Fig. 12, the boundary between the yellow buildings and the gray shadows is more distinct within the red box. In Fig. 13, the boundary between the crops (indicated in green) and the bare soil (indicated in yellow) is clearly defined. In Fig. 14, the boundary between the commercial area (shown in gray) and the residential area (shown in pink) is clearly delineated. In Fig. 15, the boundary between the Plagioclase (shown in green) and the Margarite (shown in blue) is also clearly delineated. This effect is largely attributed to the nonlinear spatial–spectral feature extraction capability of the NWGCN module. Another advantage of the proposed method is less noise. On all four datasets, the classification is more uniform over large areas, reducing the number of noise points in small regions, which is due to the elimination of redundant information by the ID-MGR module.

Traditional methods (SVM, PCA, and NPFS) fail to fully exploit spatial continuity. GCN-based methods (SSGRN, DSR-GCN, GiGCN, EMS-GCN, and MGR-NWGCN) achieve better performance, suggesting that GCNs are effective in modeling spatial or spectral correlations. Other DL methods (AMS-M2ESL and SpectralFormer) are effective in modeling spectral sequences but may be less capable of integrating spatial information compared to GCN-based approaches.

For the PaviaU dataset shown in Fig. 12, the comparison methods fail to clearly delineate the boundaries between buildings and shadows, as well as between buildings and the ground. This is because the comparison methods neglect the consideration of spatial nonlinearity, leading to incorrect integration of local information. Additionally, traditional methods are disadvantaged by their poor redundant information suppression, resulting in a large number of noise points in the classification result map.

For the Xuzhou dataset shown in Fig. 13, the classification results of the comparison methods in the region indicated by the red box consist of a large green block. This occurs because the label map only provides the crop category shown in green, leading to excessive overfitting of the model under the strong constraint of classification loss, which results in poor generalization ability. In contrast, the proposed method correctly classifies this region, demonstrating the improvement in model generalization ability brought by the MGR module.

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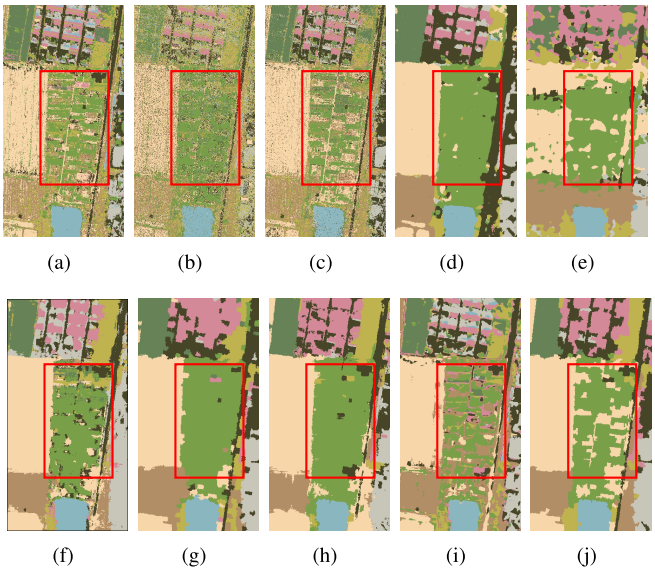


Fig. 13. Classification maps on the Xuzhou dataset. (a) SVM (80.51%). (b) PCA (61.59%). (c) NPFS (78.08%). (d) AMS-M2ESL (95.82%). (e) SSGRN (75.82%). (f) DSR-GCN (91.27%). (g) GiGCN (94.81%). (h) EMS-GCN (92.09%). (i) SpectralFormer (85.43%). (j) MGR-NWGCN (95.96%).

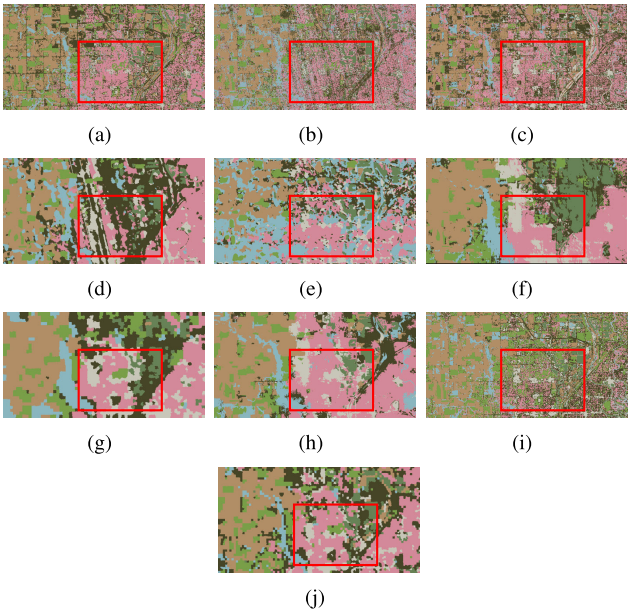


Fig. 14. Classification maps on the Kansas dataset. (a) SVM (82.60%). (b) PCA (75.89%). (c) NPFS (73.71%). (d) AMS-M2ESL (81.00%). (e) SSGRN (81.07%). (f) DSR-GCN (83.79%). (g) GiGCN (81.16%). (h) EMS-GCN (84.08%). (i) SpectralFormer (76.05%). (j) MGR-NWGCN (86.30%).

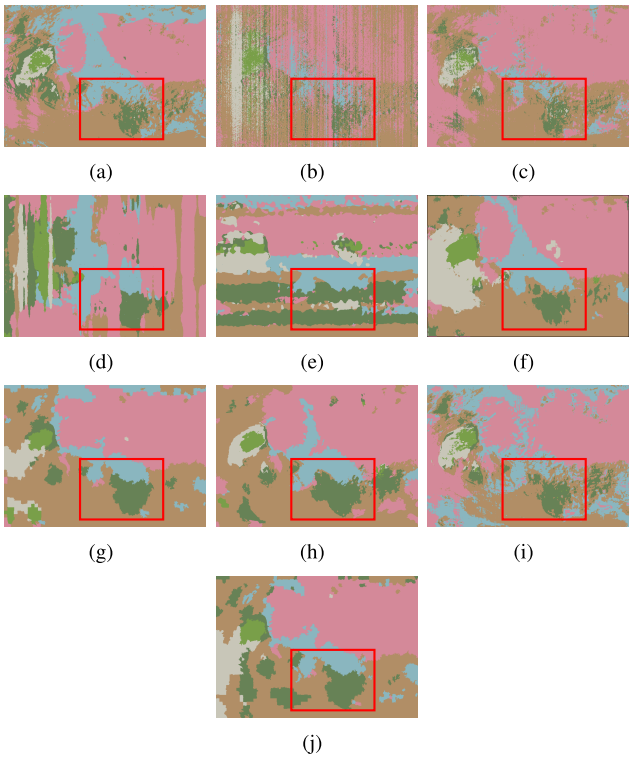


Fig. 15. Classification maps on the HC dataset. (a) SVM (87.25%). (b) PCA (71.93%). (c) NPFS (83.76%). (d) AMS-M2ESL (71.02%). (e) SSGRN (74.60%). (f) DSR-GCN (81.47%). (g) GiGCN (89.48%). (h) EMS-GCN (91.12%). (i) SpectralFormer (87.52%). (j) MGR-NWGCN (92.05%).

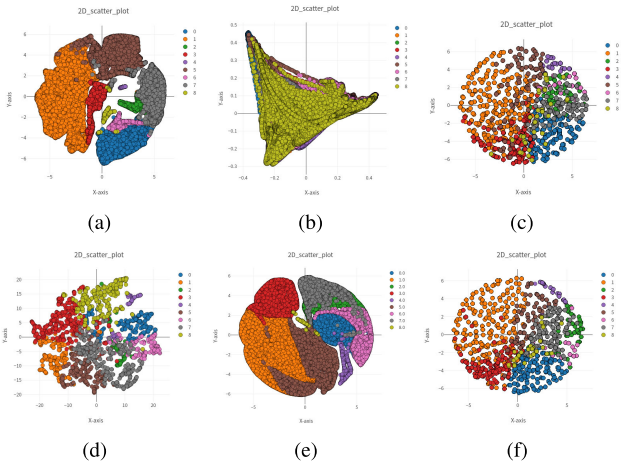


Fig. 16. t-SNE visualization of the output feature vectors produced by six methods on PaviaU dataset. (a) AMS-M2ESL. (b) DSR-GCN. (c) GiGCN (super-pixels). (d) EMS-GCN (super-pixels). (e) SpectralFormer. (f) MGR-NWGCN (super-pixels).

However, this generalization ability is not reflected in the quantitative metrics, as the region has not been annotated.

As shown in Fig. 14, on the Kansas dataset, traditional methods have the advantage of better classifying elongated categories like roads compared to DL algorithms. This is because, while DL models extract deeper abstract features, they lose some information in the process. The proposed method improves the accuracy of local neighborhood similarity measurement through the C-MGR module, reducing the interference of complex land cover categories on the central pixel, thereby enhancing the fairness of the model.

As shown in Fig. 15, in the Mars exploration task, most methods exhibit unsatisfactory performance due to the spectral similarity among surface minerals, which leads to increased interference. Under the condition of limited training samples, PCA-based dimensionality reduction methods suffer from performance degradation. Moreover, conventional approaches fail to effectively exploit spectral information, resulting in further accuracy decline. In contrast, the proposed MGR-NWGCN

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TABLE X
COMPARISONS IN TERMS OF THE PARAMETERS QUANTITIES, FLOPS, TRAINING TIME, AND TEST TIME OF DIFFERENT DL-BASED METHODS

Dataset	Complexity/Time	AMS-M2ESL	SSGRN	DSR-GCN	GIGCN	EMS-GCN	SpectralFormer	MGR-NWGCN
PaviaU	Train. (s)	28.65	204.83	147.52	35.16	42.31	17.21	480.20
	Test. (s)	0.15	0.78	1.62	0.28	0.15	5.98	0.56
	Params (K)	1053.02	30773.54	184.40	55.82	201.69	158.12	55.82
	FLOPs (M)	18.94	8730.28	612.36	2809.13	14683.99	882.92	2809.13
XuZhou	Train. (s)	51.66	167.39	189.81	35.39	30.06	29.84	367.36
	Test. (s)	0.45	0.55	1.31	0.28	0.123	7.09	0.53
	Params (K)	3756.86	12012.10	184.40	214.30	68.23	335.92	68.23
	FLOPs (M)	784.85	3607.97	387.01	3380.72	10867.95	2185.70	3380.72
Kansas	Train. (s)	109.58	268.99	123.80	62.16	45.70	29.71	600.34
	Test. (s)	0.41	0.75	1.80	0.45	0.20	6.14	1.64
	Params (K)	3756.60	34921.59	184.01	68.01	214.04	332.72	68.10
	FLOPs (M)	414.54	9850.63	537.66	5849.34	18418.35	2146.37	5849.34
HC	Train. (s)	320.22	443.26	158.48	109.58	49.29	19.39	643.95
	Test. (s)	0.51	0.93	1.92	0.78	0.19	4.79	1.11
	Params (K)	17915.99	44630.23	183.82	98.76	245.11	1257.86	98.76
	FLOPs (M)	222.99	12450.25	543.60	14109.66	28578.00	6853.39	14109.66

achieves relatively superior classification performance, benefiting from its ability to extract joint nonlinear spatial–spectral features.

3) Feature Visualization: In order to explore data distributions intuitively and reveal underlying structures and relationships, the t-SNE algorithm is applied to effectively preserve local and global structures when reducing high-dimensional data to a 3-D space, providing intuitive and valuable insights for subsequent in-depth analysis. For the feature visualization analysis, we use the output feature vectors produced by the trained MGR-NWGCN model. The same strategy is applied to the baseline methods, where their respective trained models provide the features for comparison.

As shown in Fig. 16(a), the distribution of feature after using the AMS-M2ESL method are a relatively loose distribution. The feature distribution in Fig. 16(b) is difficult to separate with a clear decision boundary, making it challenging to distinguish between different categories in the embedded space. The distribution of features using SpectralFormer in Fig. 16(e) exhibits severe overlapping. The GiGCN and EMS-GCN methods with super-pixels in Fig. 16(c) and (d) show an improvement in clustering. However, the distribution of features from different categories still overlaps, leading to some degree of mixed distributions between different classes. The GR-NWGCM method shows the most defined clustering effect in Fig. 16(f), with feature points more tightly grouped, further improving the separability of the features in the embedded space.

4) Model Complexity Analysis: To achieve high classification accuracy in the HSIC task while ensuring deployability, it is essential to conduct a quantitative analysis of model complexity. Key indicators such as floating point operations (FLOPs), parameter quantities, average training time, and average test time are critical for evaluating the computational cost of DL methods [81]. To this end, we compare the proposed MGR-NWGCN method with several DL-based methods across four datasets. The results are summarized in Table X. AMS-M2ESL and SSGRN exhibit extremely high parameter quantities, and EMS-GCN and SSGRN exhibit extremely high FLOPs, leading to high complexity and poor deployability. DSR-GCN, GIGCN, and SpectralFormer achieve moderate performance in training and test times, but with some accuracy degradation. Although the training time of the MGR-NWGCN method is relatively long, the inference speed is satisfactory. Across all datasets, MGR-NWGCN consistently demonstrates lower test time and reasonable parameter counts, highlighting its potential for practical deployment.

V. CONCLUSION

In this article, an MGR-NWGCN is proposed to obtain more accurate feature representations for the HSIC task. The MGR-NWGCN model consists of three modules, namely, NWGCN, C-MGR, and ID-MGR, as well as a regularization strategy called the MGR strategy. By adopting a graph-in-graph structure based on a cosine similarity-based adjacency matrix, the NWGCN module achieves nonlinear spatial–spectral joint information extraction. Then, to improve the accuracy of manifold surface measurement, the C-MGR module is implemented, which improves the generalization ability of the model. Meanwhile, to reduce the interference of redundant information on feature representation, the ID-MGR module is developed, which constrains the model to embed noise into low-dimensional manifold surfaces through ID estimation. Finally, the MGR strategy ensures that during model training, the cross-entropy loss, C-MGR, and ID-MGR are of the same order of magnitude. Extensive experiments are conducted on four datasets, and the proposed MGR-GCN method is compared with nine methods. The experimental results show that MGR-GCN outperforms the competing methods, indicating the effectiveness and robustness of the proposed model.

In future work, we plan to extend the proposed MGR-GCN method to address cross-scene HSIC tasks.

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